

**the latest version of java is 25 which was released in 2025

**control statement

1.Branchng

1.if

- 1.if,if.. else if
- 2.switch
- 3.conditional statement or ternary statement(?:)

2.Looping

- 1.for 2.while 3.do while

Source code

```
import java.util.Scanner;
class prg7{
public static void main(String[] args){
Scanner sc= new Scanner (System.in);
int n=sc.nextInt();
if (n<0){
System.out.println("negative number:"+n);
}
if(n>=0 && n<10){
System.out.println("Single digit: "+n);
}
else if(n>=10 && n<100){
System.out.println("Double digit: "+n);
}
else if(n>=100 && n<1000){
System.out.println("Triple digit: "+n);
}
else{
System.out.println("More than three digits: "+n);
}
}}
```

ASCII is American Standard Code for
Information Interchange

Every keys has an ascii code

- Capital(65-90)
- small(97-122)
- number(48-57)
- null(0)
- Non printable characters -1 to 31
(shift ,ctrl,up/down/left/right arrow
,capslock ,tab,function key, etc...)
- Space-32
- Symbols-33 to 47,58 to 64,91 to
96,123 to 126
- From 128- reserved extended ascii values.
- Total ascii-256

cook@pop-os:~\$ ascii -d

0 NUL	16 DLE	32	48 0	64 @	80 P	96 `	112 p
1 SOH	17 DC1	33 !	49 1	65 A	81 Q	97 a	113 q
2 STX	18 DC2	34 "	50 2	66 B	82 R	98 b	114 r
3 ETX	19 DC3	35 #	51 3	67 C	83 S	99 c	115 s
4 EOT	20 DC4	36 \$	52 4	68 D	84 T	100 d	116 t
5 ENQ	21 NAK	37 %	53 5	69 E	85 U	101 e	117 u
6 ACK	22 SYN	38 &	54 6	70 F	86 V	102 f	118 v
7 BEL	23 ETB	39 '	55 7	71 G	87 W	103 g	119 w
8 BS	24 CAN	40 (	56 8	72 H	88 X	104 h	120 x
9 HT	25 EM	41)	57 9	73 I	89 Y	105 i	121 y
10 LF	26 SUB	42 *	58 :	74 J	90 Z	106 j	122 z
11 VT	27 ESC	43 +	59 ;	75 K	91 [	107 k	123 {
12 FF	28 FS	44 ,	60 <	76 L	92 \	108 l	124
13 CR	29 GS	45 -	61 =	77 M	93]	109 m	125 }
14 SO	30 RS	46 .	62 >	78 N	94 ^	110 n	126 ~
15 SI	31 US	47 /	63 ?	79 O	95 _	111 o	127 DEL

Getting input as char

Code 1:

```
char c=sc.next().charAt(0);//to get input a single character
int as=c;//for printing the ascii number for given character
System.out.println("Character : "+c+" ASCII number :"+as);
```

Code 2:

```
if(as>=65 && as<=90){
    System.out.println("Capital letter");
}
else if(as>=97 && as<=122){
    System.out.println("Small letter");
}
else if(as>=48 && as<=57){
    System.out.println("Number");
}
else if((as>=33 && as<=47)|| (as>=58 && as<=64)|| (as>=91 && as<=96)|| (as>=123 &&
as<=126)){
    System.out.println("Symbol");
}
```

Code 3: **Printing which num is big**

```
if (a>b){
    if (a>c){
        System.out.println("a is big");
    }else{
        System.out.println("c is big");
    }
}
else{
    if(b>c){
        System.out.println("b is big");
    }
    else{
        System.out.println("c is big");
    }
}

(or)

System.out.println((a>b)?((a>c)?"a is big":"c is big"):(b>c?"b is big":"c is big"));
```

Check whether it is a vowel or not

Code :

```
char ch =sc.next().charAt(0);
switch(ch){
    case 'a':
    case 'e':
    case 'i':
    case 'o':
    case 'u':
    case 'A':
    case 'E':
    case 'I':
    case 'O':
```

```
case 'U':  
    System.out.println("Vowel : "+ch);  
    break;  
default:  
    System.out.println("Not vowel: "+ch);
```

If Vs Switch

- >if checks the condition at the form of true or false
- >switch choose options (can choose multiple options .that's why using break)
- >can use default anywhere in the switch

$n\%5==0$ & $n\%11==0$ //**divisible by 5 and 11**

Math.abs(variable/value) //for printing the absolute number (without negative sign)